

SingleRAN

Uplink Flexible Spectrum Scheduling Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 04 (2025-07-30).

- Technical changes

Change Description	Parameter Change	Base Station Model
Added support for a 0-μs interval between two consecutive switches in FSA scenarios. For details, see: • 4.1.1 Carrier Management • 4.3.6 Others	None	3900 and 5900 series base stations (macro base stations)
Optimized the carrier combination switching when no SUL PHR is reported in 1 TDD+2 FDD FSA scenarios. For details, see: • 4.1.1 Carrier Management • 4.2.2 Impacts • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Added the NRDUCellCarrMgmt.FlexSpectAcsSulPhrCarrierIntvl parameter.	3900 and 5900 series base stations (macro base stations)

- Editorial changes
Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial

features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Applicable RAT

This document applies to NR.

2.3 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FOFD-081260	Uplink Multi-band Flexible Scheduling (Trial)	4 FSA

NOTE

Uplink Multi-band Flexible Scheduling is a trial feature in a general sense. However, it is for commercial use in certain scenarios. Specifically:

1 TDD+2 FDD FSA

2.4 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
FSA	Supported in SA networking as well as NSA and SA hybrid networking, but not in NSA networking	4 FSA

Differences between high frequency bands and low frequency bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
FSA	Supported only in low frequency bands	4 FSA

3 Overview

3.1 Background and Definition

3GPP releases earlier than 3GPP Release 18 have the following restrictions on how uplink spectrum can be utilized:

- Due to limitations in the number of UE transmit channels and transmit power, a maximum of two bands can be configured for a UE in the uplink.
- When 5G networks evolve from 1 TDD +1 FDD bands to x TDD+ y FDD ($x > 1$, $y > 1$) bands, and two bands that provide the best user experience are selected from the available bands, the carrier change must be implemented through RRC reconfiguration in accordance with 3GPP specifications. However, this type of carrier change causes a long delay, affecting uplink user experience.

To address the preceding restrictions, 3GPP Release 18 introduces the flexible spectrum access (FSA) function to enhance supplementary uplink (SUL)/uplink time division multiplexing (TDM) carrier aggregation (CA). **FSA allows a gNodeB to fully configure carriers for a UE based on the FSA band combinations supported by the UE and accurately and promptly select the optimal carrier combination through downlink control information (DCI) based on the actual transmission capability and TDD slot configuration, thus enabling uplink data transmission in all slots. In this way, dynamic and intelligent carrier selection can be achieved at the TTI level, thus improving uplink user experience.**

3.2 Band Combinations

The supported FSA band combinations are 2 TDD+1 FDD and 1 TDD+2 FDD.

- In 2 TDD+1 FDD FSA, two NR TDD frequency bands and one NR FDD frequency band are involved, and the band combination can be expressed as NR TDD1+NR TDD2+NR FDD. The carrier in the NR FDD frequency band serves as the SUL carrier, which is set up in an NR TDD1 cell and works in SUL and NR FDD co-deployment mode.
- In 1 TDD+2 FDD FSA, one NR TDD frequency band and two NR FDD frequency bands are involved, and the band combination can be expressed as NR TDD+NR FDD+NR FDD (SUL). The carrier in one NR FDD frequency band

serves as the SUL carrier, which is set up in an NR TDD cell and works in SUL and NR FDD co-deployment mode.

For details about the mapping between frequency bands and frequencies, see section 5.2 "Operating bands" in 3GPP TS 38.104 V15.5.0. The bandwidth requirement of each frequency band is described in [4.3.5 Cells](#).

2 TDD+1 FDD FSA Band Combinations

[Table 3-1](#) describes the 2 TDD+1 FDD FSA band combinations.

Table 3-1 2 TDD+1 FDD FSA band combinations

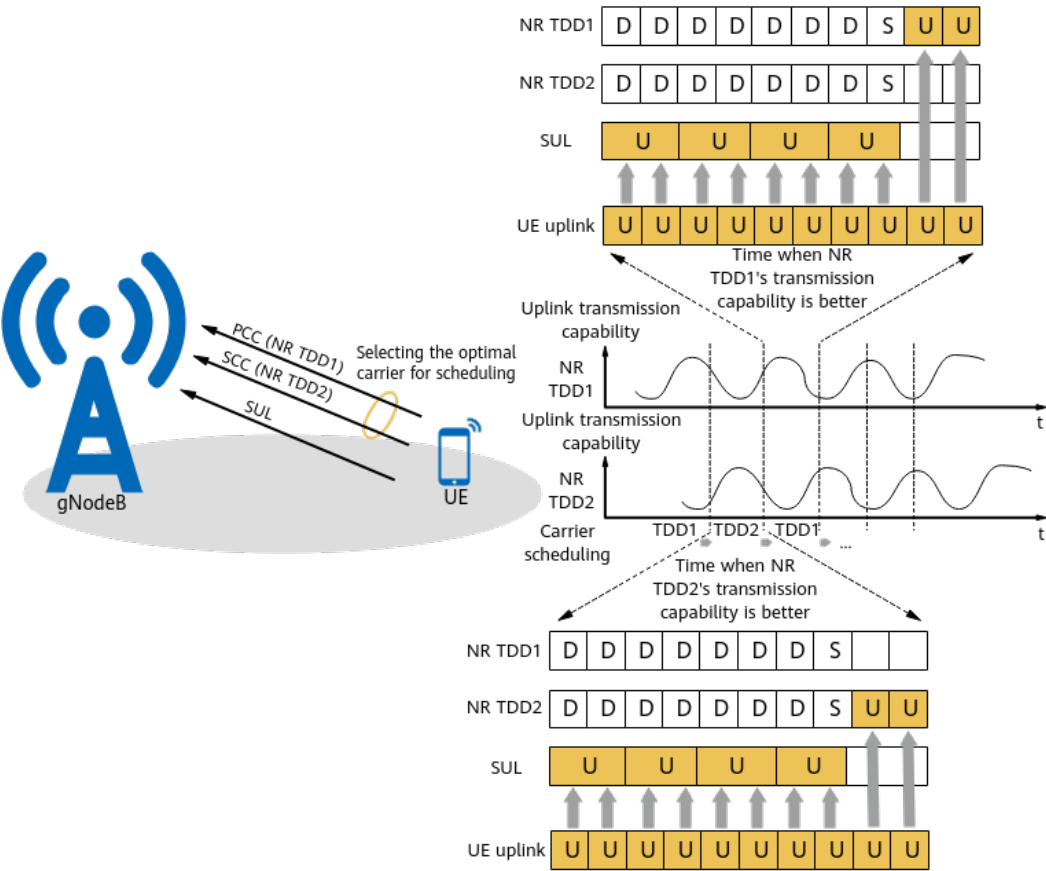
Scenario Classification	NR TDD1 Frequency Band	NR TDD1 (PCC) Slot Configuration	SUL Frequency Band	NR FDD Frequency Band Corresponding to SUL	NR TDD2 Frequency Band	NR TDD2 Slot Configuration
NR TDD1+NR TDD2+NR FDD	n78	Single-period 8:2	n81	n8	n40	Single-period 8:2
					n41	
			n83	n28	n40	
			n84	n1	n41	
			n84	n1	n40	

 NOTE

- Only one carrier is supported on the NR FDD/SUL frequency band.
- In FSA band combinations, only a carrier in the NR TDD frequency band can serve as the PCC for CA.

When the NR TDD carriers have the same slot configuration, the gNodeB selects the optimal NR TDD carrier, which together with the SUL carrier forms the optimal carrier combination. Data is transmitted using the optimal carrier combination in TDM mode. [Figure 3-1](#) shows the principle for PUSCH scheduling for 2 TDD+1 FDD three-band 3CC FSA based on the assumption that the NR TDD carriers have the same slot configuration.

Figure 3-1 Principle for PUSCH scheduling for 2 TDD+1 FDD three-band 3CC FSA when the NR TDD carriers have the same slot configuration (single-period 8:2)



One or two carriers can be configured in the NR TDD band. When two carriers are configured in the NR TDD band, intra-band contiguous uplink CA works between them, and therefore they must be served by the same gNodeB.

- When only one carrier is configured in each NR TDD band, 2 TDD+1 FDD three-band 3CC FSA is used.
- When two carriers are configured in one NR TDD band and contiguous 2CC CA works between these two carriers, 2 TDD+1 FDD three-band 4CC FSA is used.
- When two carriers are configured in each NR TDD band and contiguous 2CC CA works between these two carriers, 2 TDD+1 FDD three-band 5CC FSA is used.

1 TDD+2 FDD FSA Band Combinations

Table 3-2 describes the 1 TDD+2 FDD FSA band combinations.

Table 3-2 1 TDD+2 FDD FSA band combinations

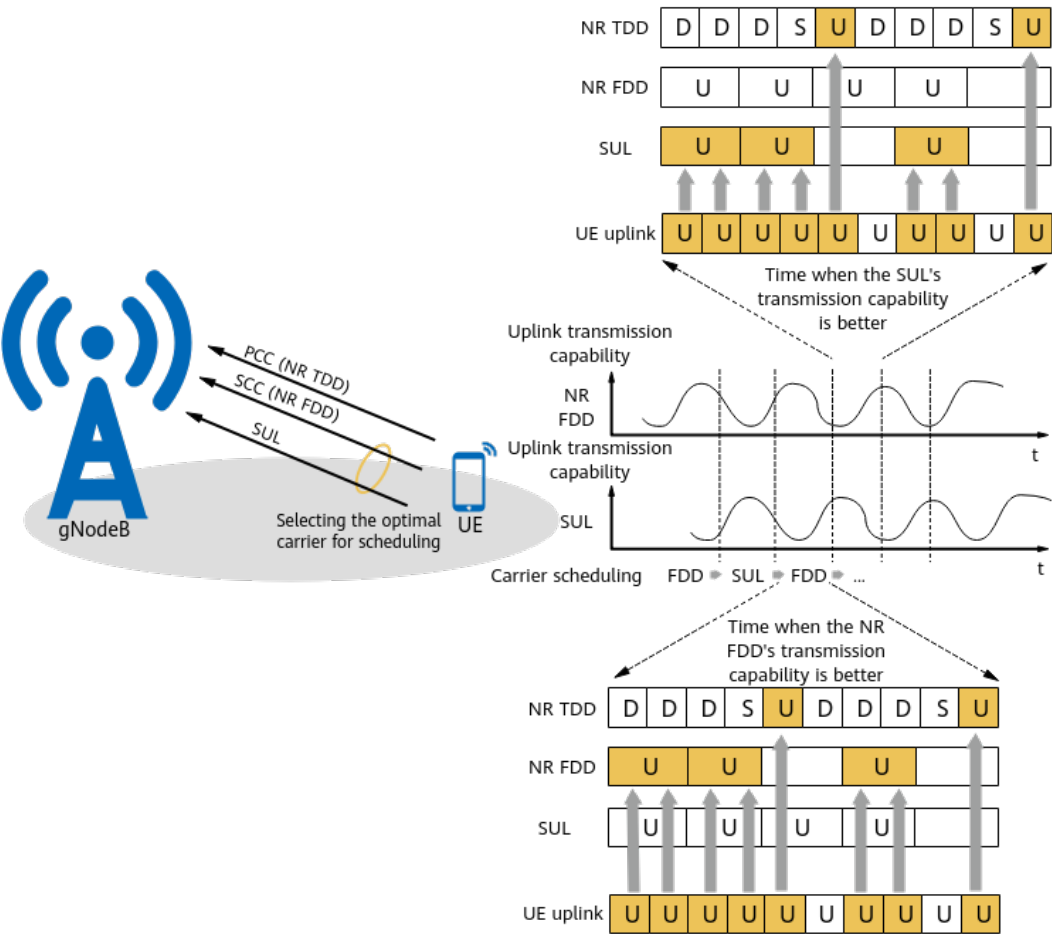
Scenario Classification	NR TDD1 Frequency Band	NR TDD1 (PCC) Slot Configuration	SUL Frequency Band	NR FDD Frequency Band Corresponding to SUL	NR FDD Frequency Band
NR TDD+NR FDD+NR FDD (SUL)	n78	4:1	n83	n28	n1
			n84	n1	n28

NOTE

- Only one carrier is supported on the NR FDD/SUL frequency band.
- In 1 TDD+2 FDD FSA, only an NR TDD carrier can serve as the PCC.

The gNodeB selects the optimal NR FDD carrier, which together with the NR TDD carrier forms the optimal carrier combination. Data is transmitted using the optimal carrier combination in TDM mode. Figure 3-2 shows the principle for PUSCH scheduling for 1 TDD+2 FDD three-band 3CC FSA.

Figure 3-2 Principle for PUSCH scheduling for 1 TDD+2 FDD three-band 3CC FSA (PCC slot configuration: 4:1)



One or two carriers can be configured in the NR TDD band. When two carriers are configured in the NR TDD band, intra-band contiguous uplink CA works between them, and therefore they must be served by the same gNodeB.

- When only one carrier is configured in the NR TDD band, 1 TDD+2 FDD three-band 3CC FSA is used.
- When two carriers are configured in the NR TDD band and contiguous 2CC CA works between these two carriers, 1 TDD+2 FDD three-band 4CC FSA is used.

4 FSA

4.1 Principles

3GPP releases earlier than 3GPP Release 18 stipulate that a maximum of two bands can be configured for a UE in the uplink. When 5G networks evolve from 1 TDD+1 FDD bands to x TDD+ y FDD ($x > 1$, $y > 1$) bands, 3GPP specifications require that RRC reconfiguration should be used to select two bands from the available bands to provide the best user experience. However, this type of carrier change causes a long delay, and uplink user experience is affected.

In this context, 3GPP Release 18 introduces the FSA function. FSA allows carriers to be configured in full mode for a UE. FSA also helps accurately and promptly select the optimal carrier combination through DCI based on the actual transmission capability and TDD frame configuration, allowing for uplink data transmission in all slots. In this way, dynamic and intelligent carrier selection can be achieved at the TTI level, improving uplink user experience.

FSA introduces enhancements to SUL and differs from "super uplink with SUL and NR FDD co-deployment" mainly in carrier management, uplink scheduling, and timing advance. For details about the principles of super uplink, see *Super Uplink*.

To enable FSA, you need to select the **FLEX_SPCT_ACCESS_SW** option of the **NRDUCellCarrMgmt.MultiBandFlexSchSwitch** parameter for both NR TDD and NR FDD cells.

You will also need to select the **FSA_CARR_SELECT_SW** option of the **NRCelAlgoSwitch.MultiFreqAlgoSwitch** parameter for the NR TDD cell if the scenario is either of the following:

- 2 TDD+1 FDD FSA, and the NR TDD carriers having the same slot configuration
- 1 TDD+2 FDD FSA

4.1.1 Carrier Management

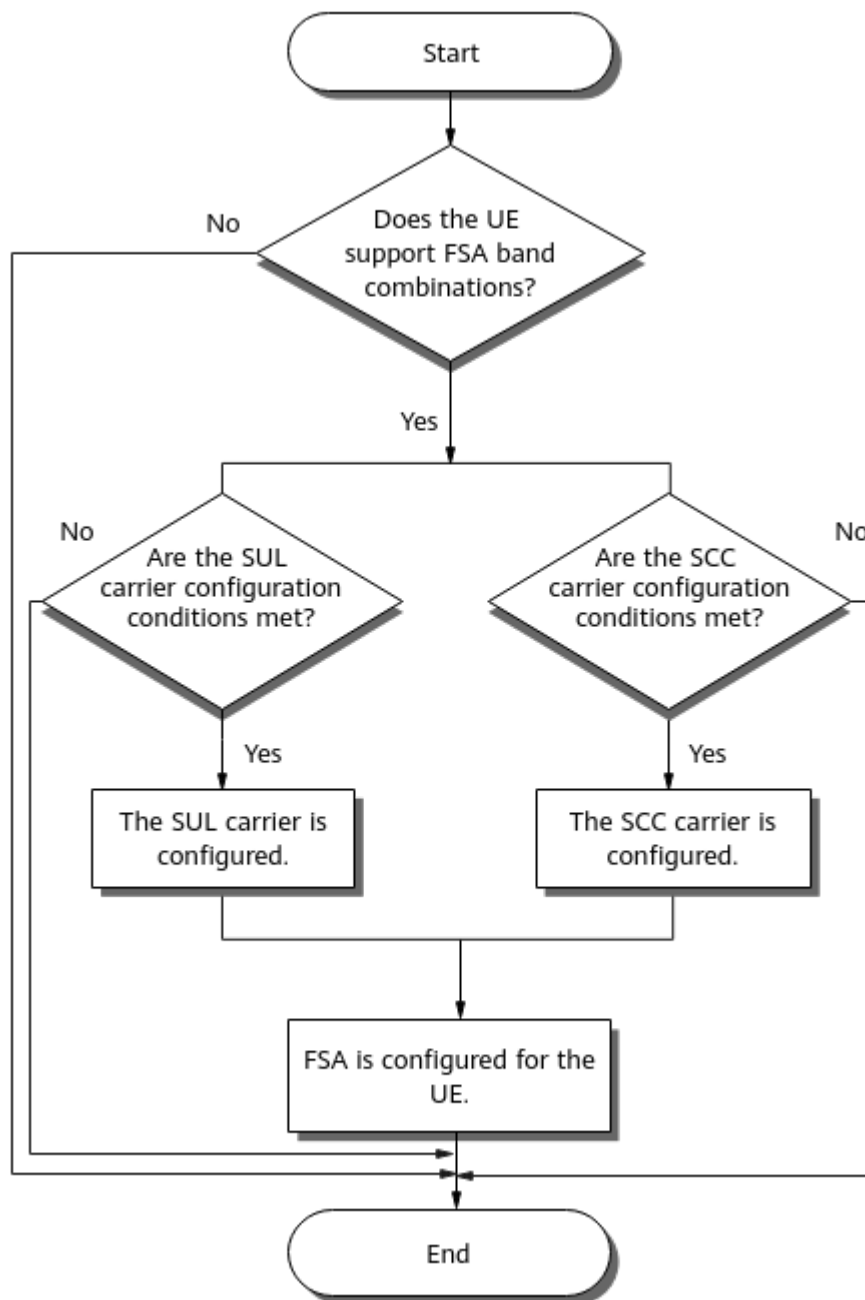
This section describes FSA carrier configuration, change, and removal.

FSA Carrier Configuration

A gNodeB starts FSA carrier configuration for a UE during initial access, an incoming handover, an incoming RRC connection reestablishment, or an RRC resume procedure.

If the UE supports FSA band combinations and meets the SUL and SCC carrier configuration conditions, the gNodeB configures the UE as an FSA UE, as shown in [Figure 4-1](#).

Figure 4-1 FSA carrier configuration process



In 2 TDD+1 FDD FSA band combinations, if the SUL carrier co-deployed with NR FDD meets the SUL carrier configuration conditions and is successfully configured,

and an NR TDD carrier meets the SCC carrier configuration conditions and is successfully configured as an SCC, the gNodeB configures the UE as an FSA UE through RRC reconfiguration.

In 1 TDD+2 FDD FSA band combinations, if the SUL carrier co-deployed with NR FDD meets the SUL carrier configuration conditions and is successfully configured, and an NR FDD carrier meets the SCC carrier configuration conditions and is successfully configured as an SCC, the gNodeB configures the UE as an FSA UE through RRC reconfiguration.

For details about the SUL carrier configuration, see *Super Uplink*.

For details about the SCC carrier configuration procedure, see descriptions of SCell configuration in *Carrier Aggregation*.

NOTE

- Currently, the inter-carrier switching delay must be less than or equal to 140 μ s. For details about the inter-carrier switching delay, see section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 V18.0.0.
- The **uplinkTxSwitchingMinimumSeparationTime-r18** field, which indicates the minimum interval between two consecutive switches, supports a value of 0 μ s or 500 μ s. For details about this field, see section 4.2.7.1 "BandCombinationList parameters" in 3GPP TS 38.306 (V18.0.0).

FSA Carrier Change

For details about the SUL carrier change procedure, see *Super Uplink*.

For details about the SCC carrier change procedure, see descriptions related to SCell change in *Carrier Aggregation*.

In 1 TDD+2 FDD FSA scenarios, if (1) an FSA UE is using the NR TDD+NR FDD carrier combination and this combination has strong capabilities, and (2) no SUL PHR is received for a period of time longer than the value of the **NRDUCellCarrMgmt.FlexSptAcsSulPhrCarrIntvl** parameter, a carrier combination switch is triggered to obtain the capabilities of the NR TDD+SUL carrier combination for assessing the SUL carrier performance. If the NR TDD+SUL carrier combination has stronger capabilities than the NR TDD+NR FDD carrier combination, the UE uses the NR TDD+SUL carrier combination. Otherwise, the UE switches back to the NR TDD+NR FDD carrier combination.

FSA Carrier Removal

In the inter-gNodeB deployment mode, if the actual inter-gNodeB delay between the PCC and SCC is greater than 0.5 ms, FSA needs to be disabled. For details about how to query the gNodeB DU eXn user-plane one-way delay and gNodeB DU eXn user-plane two-way delay, see *Super Uplink*.

- If the **FLEX_SPCT_ACCESS_SW** option of the **NRDUCELLCARRMGMT.MultiBandFlexSchSwitch** parameter is deselected for the PCC, FSA is disabled.
- If the **FLEX_SPCT_ACCESS_SW** option of the **NRDUCELLCARRMGMT.MultiBandFlexSchSwitch** parameter is deselected for the SCC, FSA is disabled. If the number of remaining carriers meets the conditions for FSA to take effect, the gNodeB reconfigures FSA for the UE. For

details about FSA carrier configuration, see [FSA Carrier Configuration](#) in [4.1.1 Carrier Management](#).

For a 2 TDD+1 FDD UE:

- When the SUL carrier is removed, FSA is disabled for the UE, and the UE works as a single-carrier UE.
- When an SCC carrier is removed:
 - In three-band 3CC scenarios, FSA is disabled for the UE, and the UE works as a super uplink UE.
 - In three-band 4CC scenarios, the UE works as a three-band 3CC FSA UE, uplink CA+super uplink UE, or super uplink UE.
 - In three-band 5CC scenarios, the UE works as a three-band 4CC FSA UE.

For details about the uplink CA+SUL band combinations, see the frequency band combinations for the simultaneous use of super uplink and uplink CA in *Super Uplink*.

For a 1 TDD+2 FDD UE:

- When the SUL carrier is removed:
 - If the **INTRA_FR_INTER_BAND_UL_CA_ENH** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter is selected for the PCC and SCC before the SUL carrier is removed, FSA is disabled for the UE, and the UE works as an NR TDD+NR FDD TDM CA UE.
 - In other cases, FSA is disabled for the UE, and the UE works as a single-carrier UE.
- When the SCC carrier is removed, FSA is disabled for the UE and the UE works as a super uplink UE.

For details about SUL carrier removal, see *Super Uplink*.

For details about SCC carrier removal, see descriptions related to SCell removal in *Carrier Aggregation*.

Optimized Fallback upon Uplink CA Exceptions

In FSA scenarios, if an uplink SCell addition exception occurs, the gNodeB performs fallback by removing the involved uplink SCell. In some scenarios, the gNodeB may also remove the SCells for downlink CA, causing downlink CA to become ineffective as well.

To address this issue, the 'optimized fallback upon uplink CA exceptions' function has been introduced. This function is controlled by the **UL_CA_FALLBACK_DL_ONLY_CA_SW** option of the **gNodeBCarrMgmt.CaEnhancedAlgoSwitch** parameter. If this option is selected and an exception occurs in uplink SCell, the gNodeB removes only the involved uplink SCell but not downlink SCells, ensuring that downlink CA remains unaffected.

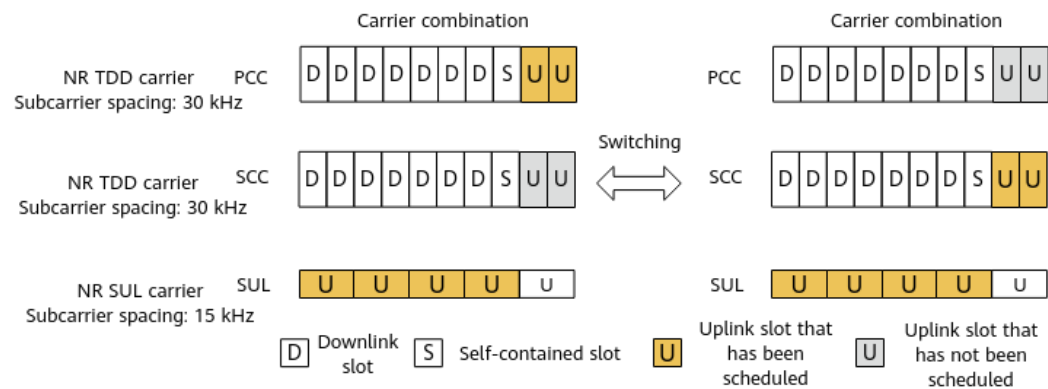
4.1.2 Uplink Scheduling

An FSA UE's uplink carriers consist of PCC, SCC, and SUL carriers.

When FSA takes effect on a UE, the optimal carrier combination is selected for uplink data transmission, and the uplink data is transmitted in the involved frequency bands in TDM mode.

Figure 4-2 shows the uplink TDM scheduling when PCC and SCC have the same slot configuration in 2 TDD+1 FDD three-band 3CC FSA scenarios, and the benefits in such scenarios come from optimal carrier selection. For details about the benefits, see **4.2.1 Benefits**.

Figure 4-2 Uplink TDM scheduling for an FSA UE when the PCC and SCC use the single-period 8:2 slot configuration

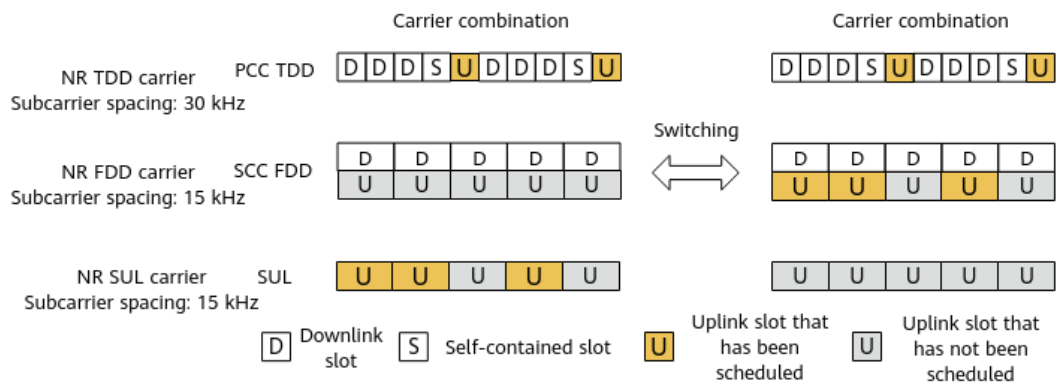


After 2 TDD+1 FDD FSA takes effect on a UE, the UE's uplink data is transmitted on the SUL carrier in the downlink slots of the NR TDD carrier and in the downlink symbols of the self-contained slots and is transmitted in the uplink slots of the NR TDD carrier.

- The scheduling information for the uplink PCC or SUL carrier is delivered through DCI on the PCC, and the UL/SUL indicator in the DCI indicates whether the carrier involved is an NR TDD or SUL carrier. For details, see section 7.3 "Downlink control information" in 3GPP TS 38.212 (V18.2.0).
- The scheduling information for the uplink SCC is delivered through DCI on the SCC.

Figure 4-3 shows uplink data scheduling for a 1 TDD+2 FDD three-band 3CC FSA UE in TDM mode.

Figure 4-3 Uplink TDM scheduling for the FSA UE when the PCC uses the 4:1 slot configuration



After 1 TDD+2 FDD FSA takes effect on a UE, the UE's uplink data is transmitted on the SUL or NR FDD carrier in the downlink slots of the NR TDD carrier and in the downlink symbols of the self-contained slots and is transmitted in the uplink slots of the NR TDD carrier.

- The scheduling information for the uplink PCC or SUL carrier is delivered through DCI on the PCC, and the UL/SUL indicator in the DCI indicates whether the carrier involved is an NR TDD or SUL carrier. For details, see section 7.3 "Downlink control information" in 3GPP TS 38.212 (V18.2.0).
- The scheduling information for the uplink FDD SCC is delivered through DCI on the FDD SCC.

FSA Carrier Combination Management

The gNodeB selects the optimal carrier combination for transmission based on the transmission capabilities of each carrier combination.

For carrier combination switching, the uplink transmission capability of the target carrier combination must be greater than or equal to that of the source carrier combination multiplied by $(1 + \text{NRDUCellCarrMgmt.FlexSpctAccessExpEvalHyst})$.

CAUTION

The minimum interval between two carrier combination switches is 20 ms.

4.1.3 Timing Advance

FSA supports only a single timing advance group (TAG). That is, the PCC, SCC, and SUL share the same set of uplink timing mechanisms, and M-TAG is not supported.

4.2 Network Analysis

4.2.1 Benefits

FSA helps improve uplink user experience, but it does so only for one FSA UE and only when certain other requirements are met. When there are multiple FSA UEs, such user experience improvements cannot be guaranteed.

For a single 2 TDD+1 FDD FSA UE, the uplink data rate is expected to improve by 2% to 3% when the NR TDD carriers use the same slot configuration.

For a single 1 TDD+2 FDD FSA UE, the uplink data rate is expected to improve by 5% to 20%.

However, FSA gains can be negatively affected by the following factors:

- Factors such as inter-gNodeB time synchronization deviation and different site locations result in a difference in TA between NR TDD and SUL carriers, which in turn results in a decrease in SUL performance gains and consequently a decrease in peak throughput gains.
- The smaller the fluctuation in the load difference (such as the **Uplink Resource Block Utilizing Rate (DU)**) between carriers, the smaller the FSA gains.
- The larger the inter-carrier switching delay (uplinkTxSwitchingPeriod) for FSA carriers, the smaller the FSA gains.
- A smaller uplink traffic volume of the FSA UE may also lead to smaller FSA gains.
- FSA gains are affected by the UE's sounding reference signal (SRS) period. A smaller cell load (indicated by **N.User.RRCCConn.Avg**) leads to a shorter SRS period and consequently smaller FSA gains.
- If **NRDUCellPusch.SchMultiSlivTypeUeSwitch** is turned off before FSA is enabled, the average uplink UE throughput may decrease after FSA is enabled.
- To ensure the prompt switching of carrier combinations and enhance the accuracy of data volume distributed to SCells in the uplink, it is recommended that the **MF_ACTUAL_PUSCH_ILPC_SW** option of the **NRDUCellULPcConfig.UlPwrCtrlAlgoExtSwitch** parameter be selected to enable multi-frequency actual-PUSCH-based inner-loop power control. For details about this function, see *Power Control*.

4.2.2 Impacts

For details about the impact between the function and other functions, see [4.3.2.3 Function Impacts](#).

After FSA is enabled, more cell configuration information needs to be exchanged over the eXn interface between gNodeBs. This results in an increase in the values of counters related to the received and transmitted traffic volumes in the **SCTP Link Measurement** function subset.

In 1 TDD+2 FDD FSA scenarios, a smaller value of the **NRDUCellCarrMgmt.FlexSpectAcsSulPhrCarrIntvl** parameter results in more carrier combination changes triggered by SUL PHR guarantee for FSA UEs. In addition, the peak rate of these UEs will slightly fluctuate. A larger value of this parameter results in fewer carrier combination changes triggered by SUL PHR guarantee for FSA UEs.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-081260	Uplink Multi-band Flexible Scheduling (Trial)	NR0S00UMFS00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists* in 5G RAN feature documentation.

4.3.2 Software

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Super uplink	FLEX_SUPER_UP LINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter INTER_GNB_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter selected for both NR TDD and NR FDD cells when the cells are served by different gNodeBs	<i>Super Uplink</i>	When FSA is enabled, inter-base-station flexible super uplink must be enabled in the NR TDD cell that serves as the PCell and the NR FDD cell that serves as the SUL cell (SUL and NR FDD co-deployment).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Intra-FR1 multi-carrier uplink CA	FR1_UL_CA_MULTI_CC_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	None	When FSA is enabled: • In 2 TDD+1 FDD three-band 4CC and 5CC scenarios, the FR1_UL_CA_MULTI_CC_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter must be selected for the NR TDD cell that serves as the PCell. • In 1 TDD+2 FDD three-band 4CC scenarios, the FR1_UL_CA_MULTI_CC_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter must be selected for the NR TDD cell that serves as the PCell. • With the FR1_UL_CA_MULTI_CC_SW option selected for the PCell, 1 TDD+2 FDD three-band 4CC FSA can take effect only when the base station that serves the SCC/SUL runs V100R021C00 or a later version.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Uplink intra-gNodeB intra-band CA	INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	In NR TDD intra-band contiguous CA scenarios, when FSA is enabled, the INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter must be selected for cells involved in intra-band contiguous CA.
Low-frequency NR TDD	Downlink intra-gNodeB intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When FSA is enabled, the PCC and SCC require downlink intra-gNodeB intra-FR inter-band CA.
Low-frequency NR TDD	Downlink inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	When FSA is enabled on NR TDD carriers served by different gNodeBs, downlink inter-gNodeB CA must be enabled for both the PCell and SCell.

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	High-speed cells do not support FSA.
Low-frequency NR TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	FSA is not supported in hyper cell networking.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransmissionMode set to JOINT_MODE	<i>Cell Combination</i>	FSA is not supported in Joint Transmission Cell Combination networking.
Low-frequency NR TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	FSA is not supported in Cell Combination networking.
Low-frequency NR TDD	Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	A cell enabled with Fusion Cell does not support FSA.
Low-frequency NR TDD	Positioning-dedicated SRS transmission in uplink slots	SRS_FOR_POSITION_ON_USLOT_SW option of the NRDUCellPosAlgo.LcsMeasureAlgoSw parameter	<i>Positioning</i>	FSA cannot be enabled together with "positioning-dedicated SRS transmission in uplink slots".
Low-frequency NR TDD	RRC reconfiguration procedure optimization	RRC_RECFG_PROCESS_OPT_SW option of the gNodeBParam.ProcessSwitch parameter	None	FSA cannot be enabled together with RRC reconfiguration procedure optimization.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency NR TDD	SRS Carrier Switching Time UL Threshold	NRDUCellCarrMgmt.SrsCarrSwitchingTimeULThld	None	When FSA is enabled, this parameter can only be set to N140 .
NR FDD	UL Channel Estimation Type	NRDUCellPusch.UlChnEstimationType set to ENHANCED	None	If the NRDUCellPusch.UlChnEstimationType parameter is set to ENHANCED for a cell, the cell does not support FSA.
Low-frequency NR TDD	Wideband SRS measurement collection	WIDEBAND_SRS_COLLECT_SW option of the NRCellFeatureSwitch.CellMeasCollectSwitch parameter	None	FSA is mutually exclusive with wideband SRS measurement collection.

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
Low-frequency NR TDD	3GPP R15-compliant uplink slot aggregation	VONR_UL_SLOT_AGGREGATION_SW option of the NRDUCellULSch . VoiceULSchSwitch parameter	<i>VoNR</i>	3GPP R15-compliant uplink slot aggregation does not take effect on UEs with FSA in effect.
Low-frequency NR TDD	3GPP R16-compliant uplink slot aggregation	VOICE_R16_UL_SLOT_AGG_SW option of the NRDUCellULSch . VoiceULSchSwitch parameter	<i>VoNR</i>	3GPP R16-compliant uplink slot aggregation does not take effect on UEs with FSA in effect.
Low-frequency NR TDD	PUSCH scheduling based on SRS interference avoidance	SRS_INTRF_AVOID_PUSCH_SWITCH option of the NRDUCellSrs . SrsAlgoExtSwitch parameter	None	PUSCH scheduling based on SRS interference avoidance does not take effect on UEs with FSA in effect.
Low-frequency NR TDD	PUSCH scheduling in self-contained slots	PUSCH_SCH_IN_S_SLOT_SW option of the NRDUCellPusch . ULPuschAlgoSwitch parameter	None	PUSCH scheduling in self-contained slots does not take effect on UEs with FSA in effect.
Low-frequency NR TDD	FWA user identification	gNBQciBearer.ServiceType set to FWA_INTERNET , FWA_VIDEO , or FWA_PRIVATE_LINE	<i>FWA</i>	FSA does not take effect on UEs with FWA in effect.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Cloud X user identification	gNBQciBearer.ServiceType set to EMBB_CLOUD_VR or EMBB_CLOUD_GAMING	<i>Ultimate Cloud X Experience</i>	Cloud X user identification does not take effect on UEs with FSA in effect.
Low-frequency NR TDD	Aperiodic CSI	None	None	FSA does not support aperiodic CSI reporting for the PCC and SCC on the SCC.
NR FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	FSA does not take effect in 16T16R or 32T32R NR FDD cells with LTE FDD and NR Flash Dynamic Spectrum Sharing enabled.
NR FDD	Hybrid DSS Based on Asymmetric Bandwidth	LTE_NR_FDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	<i>LTE FDD and NR Hybrid Dynamic Spectrum Sharing</i>	FSA does not take effect in 16T16R or 32T32R NR FDD cells with Hybrid DSS Based on Asymmetric Bandwidth enabled.

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD	Proactive avoidance of remote interference	SRS_RIM_INTERF_AVOID_SW and SRS_INTERF_COORD_S_SLOT_SW options of the NRDUCellSrsResourcesAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	In 2 TDD+1 FDD three-band 3CC scenarios, if the PCC and SCC are configured with the same slot configuration, FSA is supported only when proactive avoidance of remote interference is enabled on both the PCC and SCC. If the PCC and SCC are configured with different slot assignments, FSA is supported only when proactive avoidance of remote interference is enabled on both the PCC and SCC or on only the NR TDD cell with the single-period 8:2 slot configuration. In 2 TDD+1 FDD three-band 4CC/5CC scenarios, proactive avoidance of remote interference cannot be enabled in NR TDD cells.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5910, or BBU5900A.

Boards

Main control boards: All NR-capable main control boards support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Baseband processing units:

For macro base stations:

- For NR TDD cells, the UBBPg and later boards that support NR TDD can be used.
- For NR FDD cells, the UBBPg and later boards that support NR FDD can be used.

RF Modules

[Table 4-1](#) and [Table 4-2](#) list the RF modules that support this function when macro base stations are deployed.

Table 4-1 RF modules that support 2 TDD+1 FDD FSA when macro base stations are deployed

Frequency Band	RF Module
n78 (3.5/3.7 GHz)	Modules supporting 32T32R or 64T64R
n79 (4.9 GHz)	Modules supporting 32T32R or 64T64R
n40 (2.3 GHz)	Modules supporting 32T32R or 64T64R
n41 (2.6 GHz)	Modules supporting 32T32R or 64T64R
n8/n81 (900 MHz)	NR-FDD-capable modules
n28/n83 (700 MHz)	NR-FDD-capable modules
n1/n84 (2.1 GHz)	NR-FDD-capable modules

Table 4-2 RF modules that support 1 TDD+2 FDD FSA when macro base stations are deployed

Frequency Band	RF Modules
n78 (3.5/3.7 GHz)	Modules supporting 32T32R or 64T64R
n1/n84 (2.1 GHz)	NR-FDD-capable modules
n3/n80 (1.8 GHz)	NR-FDD-capable modules
n5/n89 (800 MHz)	NR-FDD-capable modules
n8/n81 (900 MHz)	NR-FDD-capable modules
n28/n83 (700 MHz)	NR-FDD-capable modules

4.3.4 Networking

For details about the networking requirements of SUL and NR FDD co-deployment, see descriptions of the deployment of intra-/inter-base-station super uplink in *Super Uplink*.

4.3.5 Cells

- The **NRDUCell.DuplexMode** parameter for the cells is not set to **CELL_SDL**.
- The subcarrier spacing of the NR FDD and SUL carriers is 15 kHz.
- The SUL carrier only supports the SUL and NR FDD co-deployment mode and does not support the independent SUL deployment mode.
- The maximum inter-gNodeB delay allowed between the PCC and SCC is 0.5 ms. The maximum delay allowed between the PCC and SUL carrier is configured through the **NRDUCellCollabServ.NulSulMaxInterGnbDelay** and **NRDUCellCollabCoop.NulSulMaxInterGnbDelay** parameters.
- The bandwidth requirements of each frequency band are described in the following table.

Table 4-3 Bandwidth requirements of each frequency band in 2 TDD+1 FDD FSA scenarios

Frequency Band	Bandwidth Requirement
n78 (3.5/3.7 GHz)	Same as the bandwidths supported by the RF modules. For details, see the technical specifications of the RF module concerned in <i>3900 & 5900 Series Base Station Product Documentation</i> .
n79 (4.9 GHz)	
n40 (2.3 GHz)	
n41 (2.6 GHz)	
n8/n81 (900 MHz)	10 MHz, 15 MHz, and 20 MHz
n28/n83 (700 MHz)	10 MHz, 15 MHz, 20 MHz, and 30 MHz
n1/n84 (2.1 GHz)	10 MHz, 15 MHz, 20 MHz, 30 MHz, 40 MHz, and 50 MHz

Table 4-4 Bandwidth requirements of each frequency band in 1 TDD+2 FDD FSA scenarios

Frequency Band	Bandwidth Requirement
n78 (3.5/3.7 GHz)	Same as the bandwidths supported by the RF modules. For details, see related technical specifications in <i>3900 & 5900 Series Base Station Product Documentation</i> .
n1/n84 (2.1 GHz)	10 MHz, 15 MHz, 20 MHz, 30 MHz, 40 MHz, and 50 MHz

Frequency Band	Bandwidth Requirement
n3/n80 (1.8 GHz)	10 MHz, 15 MHz, and 20 MHz
n5/n89 (800MHz)	10 MHz, 15 MHz, and 20 MHz
n8/n81 (900 MHz)	10 MHz, 15 MHz, and 20 MHz
n28/n83 (700 MHz)	10 MHz, 15 MHz, 20 MHz, and 30 MHz

4.3.6 Others

UEs support different start symbols between PUCCH scheduling and PUSCH scheduling. That is, the value of **diffSymbol** in the *mux-SR-HARQ-ACK-CSI-PUCCH-OncePerSlot* IE of the **phy-Parameters** field is **supported**.

In 2 TDD+1 FDD 4CC/5CC FSA scenarios, UEs support HARQ-ACK multiplexing in a PUSCH in multi-PUSCH scheduling scenarios. That is, the value of *mux-HARQ-ACK-withoutPUCCH-onPUSCH-r16* of the **phy-Parameters** field is **supported**.

UEs must be able to switch between carriers within 35 μ s or 140 μ s. This UE capability is indicated by the *switchingPeriodFor1T-r18* IE. For details about the switching delay, see section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 (V18.0.0).

UEs support the TDM scheduling mode for the carriers in FSA band combinations. That is, the value of **uplinkTxSwitchingOptionForBandPair-r18** is **switchedUL**. For details, see 3GPP TS 38.306 (V18.0.0) and section 4.2.7.1 "BandCombinationList parameters" in 3GPP TS 38.331 (V18.0.0).

UEs support FSA introduced in 3GPP Release 18.

- UEs support the FSA band combinations. If the *BandCombinationList-UplinkTxSwitch-v1800* IE in the UECapabilityInformation message contains the FSA band combination, UEs support the FSA band combination.
- The interval between two consecutive switches is 0 μ s or 500 μ s. The interval is indicated by the **uplinkTxSwitchingMinimumSeparationTime-r18** field, and explanations of this field can be found in section 4.2.7.1 "BandCombinationList parameters" in 3GPP TS 38.306 (V18.0.0).
- In three-band scenarios, combinations of 1T and 2T transmit antennas (such as 1T + 1T + 1T and 2T + 2T + 1T) are supported in the involved bands. The 1T or 2T capability is determined by the value of *maxNumberMIMO-LayersCB-PUSCH*. If the value is **oneLayer**, 1T is used; if the value is **twoLayers**, 2T is used. For details, see section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 (V18.0.0).

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-5 and **Table 4-6** describe the parameters used for function activation in NR TDD and NR FDD cells, respectively. This section does not describe parameters related to cell establishment.

Table 4-5 Parameters used for function activation in the NR TDD cell

Parameter Name	Parameter ID	Setting Notes
Multi-Band Flexible Scheduling Switch	NRDUCellCarrMgmt.Mul tiBandFlexSchSwitch	Select the FLEX_SPCT_ACCESS_SW option of this parameter.
SRS Carrier Switching Time UL Threshold	NRDUCellCarrMgmt.Srs CarrSwitchingTi- meULThld	Set this parameter to N140 .
Multi-Frequency Algorithm Switch	NRCellAlgoSwitch.Multi FreqAlgoSwitch	Select the FSA_CARR_SELECT_SW option if the scenario is either of the following: • 2 TDD+1 FDD FSA, and the NR TDD carriers having the same slot configuration • 1 TDD+2 FDD FSA
CA Enhanced Algorithm Switch	gNodeBCarrMgmt.CaEn hancedAlgoSwitch	Select the UL_CA_FALLBACK_DL_ONLY_CA_SW option.
Flex Spct Acs SUL PHR Carr Comb Chg Intvl	NRDUCellCarrMgmt.Fle xSpctAcsSulPhrCarrIntvl	Set this parameter to 10 in 1 TDD+2 FDD FSA scenarios.
UL Power Control Algorithm Extension Switch	NRDUCellULPcCon- fig.ULPwrCtrlAlgoExtS- witch	Select the MF_ACTUAL_PUSCH_ILPC_S W option to ensure the prompt switching of carrier combinations and enhance the accuracy of data volume distributed to SCells in the uplink.

Table 4-6 Parameter used for function activation in the NR FDD cell

Parameter Name	Parameter ID	Setting Notes
Multi-Band Flexible Scheduling Switch	NRDUCellCarrMgmt. MultiBandFlexSchS- witch	Select the FLEX_SPCT_ACCESS_SW option of this parameter.

Parameter Name	Parameter ID	Setting Notes
UL Power Control Algorithm Extension Switch	NRDUCellUIPcConfig.UlPwrCtrlAlgoExtSwitch	Select the MF_ACTUAL_PUSCH_ILPC_SW option to ensure the prompt switching of carrier combinations and enhance the accuracy of data volume distributed to SCells in the uplink.

4.4.1.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

NR TDD and NR FDD cells must be configured before running the following commands. For details, see *Cell Management*. Refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

When a carrier in the NR FDD frequency band of the FSA band combination serves as the SUL carrier, the NR TDD and NR FDD cells must be configured first. For details about NR TDD cell configuration and NR FDD cell configuration, see *Super Uplink*.

If the FSA band combinations are also involved in CA, cells for CA must be configured. For details about the configurations, see *Carrier Aggregation* and *CA Experience Improvement*.

FSA supports both intra-gNodeB and inter-gNodeB deployment. This document uses 2 TDD+1 FDD three-band 3CC FSA and 1 TDD+2 FDD three-band 3CC FSA in intra-gNodeB deployment mode as an example.

The following are examples of activation commands for 2 TDD+1 FDD three-band 3CC FSA in intra-gNodeB deployment mode. Assume the following:

- The NR TDD1 cell ID is 100 and the gNodeB ID is 750074.
- The NR TDD2 cell ID is 101 and the gNodeB ID is 750074.
- The NR FDD cell ID is 120 and the gNodeB ID is 750074.
- The NR TDD1 PCC frequency band is n41, the SUL frequency band is n83, the NR FDD frequency band is n28, and the NR TDD2 SCC frequency band is n79.

//NR TDD1 cell configuration:

//Setting SrsCarrSwitchingTimeUlThld

MOD NRDUCCELLCARRMGMT: NrDuCellId=100, SrsCarrSwitchingTimeUlThld=N140;

//Turning on FLEX_SPCT_ACCESS_SW for NR TDD1 cell

MOD NRDUCCELLCARRMGMT: NrDuCellId=100, MultiBandFlexSchSwitch=FLEX_SPCT_ACCESS_SW-1;

```

//(Optional) Turning on FSA_CARR_SELECT_SW for NR TDD1 cell when the NR TDD carriers have the same
slot configuration
MOD NRCELLALGOSWITCH: NrCellId=100, MultiFreqAlgoSwitch=FSA_CARR_SELECT_SW-1;
//(Optional) (Recommended) Enabling optimized fallback upon uplink CA exceptions when an uplink SCell
addition exception occurs, the gNodeB performs fallback and removes the downlink SCells as well
MOD GNODEBCARRMGMT: CaEnhancedAlgoSwitch=UL_CA_FALLBACK_DL_ONLY_CA_SW-1;
//(Optional) Enabling multi-frequency actual-PUSCH-based inner-loop power control to ensure the prompt
switching of carrier combinations and enhance the accuracy of data volume distributed to SCells in the
uplink
MOD NRDUCELLULPCONFIG: NrDuCellId=100, UIPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-1;
//NR TDD2 cell configuration:
//Setting SrsCarrSwitchingTimeUlThld
MOD NRDUCELLCARRMGMT: NrDuCellId=101, SrsCarrSwitchingTimeUlThld=N140;
//Turning on FLEX_SPCT_ACCESS_SW for NR TDD2 cell
MOD NRDUCELLCARRMGMT: NrDuCellId=101, MultiBandFlexSchSwitch=FLEX_SPCT_ACCESS_SW-1;
//(Optional) Enabling multi-frequency actual-PUSCH-based inner-loop power control to ensure the prompt
switching of carrier combinations and enhance the accuracy of data volume distributed to SCells in the
uplink
MOD NRDUCELLULPCONFIG: NrDuCellId=101, UIPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-1;
//NR FDD cell configuration:
//Turning on FLEX_SPCT_ACCESS_SW for NR FDD cell
MOD NRDUCELLCARRMGMT: NrDuCellId=120, MultiBandFlexSchSwitch=FLEX_SPCT_ACCESS_SW-1;
//(Optional) Enabling multi-frequency actual-PUSCH-based inner-loop power control to ensure the prompt
switching of carrier combinations and enhance the accuracy of data volume distributed to SCells in the
uplink
MOD NRDUCELLULPCONFIG: NrDuCellId=120, UIPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-1;

```

The following are examples of activation commands for 1 TDD+2 FDD three-band 3CC FSA in intra-gNodeB deployment mode. Assume the following:

- The NR DU cell ID of the NR TDD cell is 0.
- The NR DU cell ID of the NR FDD cell is 1.
- The NR DU cell ID of the SUL cell is 2.

```

//NR TDD cell configuration:
//Setting SrsCarrSwitchingTimeUlThld
MOD NRDUCELLCARRMGMT: NrDuCellId=0, SrsCarrSwitchingTimeUlThld=N140;
//Turning on FLEX_SPCT_ACCESS_SW for the NR TDD cell
MOD NRDUCELLCARRMGMT: NrDuCellId=0, MultiBandFlexSchSwitch=FLEX_SPCT_ACCESS_SW-1;
//Turning on FSA_CARR_SELECT_SW for the NR TDD cell
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=FSA_CARR_SELECT_SW-1;
//(Optional) (Recommended) Enabling optimized fallback upon uplink CA exceptions when an uplink SCell
addition exception occurs, the gNodeB performs fallback and removes the downlink SCells as well
MOD GNODEBCARRMGMT: CaEnhancedAlgoSwitch=UL_CA_FALLBACK_DL_ONLY_CA_SW-1;
//Setting FlexSptAcsSulPhrCarrIntvl
MOD NRDUCELLCARRMGMT: NrDuCellId=0, FlexSptAcsSulPhrCarrIntvl=10;
//(Optional) Enabling multi-frequency actual-PUSCH-based inner-loop power control to ensure the prompt
switching of carrier combinations and enhance the accuracy of data volume distributed to SCells in the
uplink
MOD NRDUCELLULPCONFIG: NrDuCellId=0, UIPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-1;
//NR FDD cell configuration:
//Turning on FLEX_SPCT_ACCESS_SW for the NR FDD cell
MOD NRDUCELLCARRMGMT: NrDuCellId=1, MultiBandFlexSchSwitch=FLEX_SPCT_ACCESS_SW-1;
//(Optional) Enabling multi-frequency actual-PUSCH-based inner-loop power control to ensure the prompt
switching of carrier combinations and enhance the accuracy of data volume distributed to SCells in the
uplink
MOD NRDUCELLULPCONFIG: NrDuCellId=1, UIPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-1;
//SUL cell configuration:
//Turning on FLEX_SPCT_ACCESS_SW for SUL cell
MOD NRDUCELLCARRMGMT: NrDuCellId=2, MultiBandFlexSchSwitch=FLEX_SPCT_ACCESS_SW-1;
//(Optional) Enabling multi-frequency actual-PUSCH-based inner-loop power control to ensure the prompt
switching of carrier combinations and enhance the accuracy of data volume distributed to SCells in the
uplink
MOD NRDUCELLULPCONFIG: NrDuCellId=2, UIPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-1;

```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

The deactivation command example for FSA is as follows:

Scenario: 2 TDD+1 FDD three-band 3CC FSA deployment

```
//Turning off FLEX_SPCT_ACCESS_SW for NR TDD1 cell
MOD NRDUCELLCARRMGMT: NrDuCellId=100, MultiBandFlexSchSwitch=FLEX_SPCT_ACCESS_SW-0;
//((Optional) Turning off FSA_CARR_SELECT_SW for NR TDD1 cell when the NR TDD carriers have the same
slot configuration
MOD NRCELLALGOSWITCH: NrCellId=100, MultiFreqAlgoSwitch=FSA_CARR_SELECT_SW-0;
//((Optional) Disabling optimized fallback upon uplink CA exceptions
MOD GNODEBCARRMGMT: CaEnhancedAlgoSwitch=UL_CA_FALLBACK_DL_ONLY_CA_SW-0;
//((Optional) Disabling multi-frequency actual-PUSCH-based inner-loop power control
MOD NRDUCELLULPCONFIG: NrDuCellId=100, UIPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-0;
//Turning off FLEX_SPCT_ACCESS_SW for NR TDD2 cell
MOD NRDUCELLCARRMGMT: NrDuCellId=101, MultiBandFlexSchSwitch=FLEX_SPCT_ACCESS_SW-0;
//((Optional) Disabling multi-frequency actual-PUSCH-based inner-loop power control
MOD NRDUCELLULPCONFIG: NrDuCellId=101, UIPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-0;
//Turning off FLEX_SPCT_ACCESS_SW for the NR FDD cell
MOD NRDUCELLCARRMGMT: NrDuCellId=120, MultiBandFlexSchSwitch=FLEX_SPCT_ACCESS_SW-0;
//((Optional) Disabling multi-frequency actual-PUSCH-based inner-loop power control
MOD NRDUCELLULPCONFIG: NrDuCellId=120, UIPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-0;
```

Scenario: 1 TDD+2 FDD three-band 3CC FSA deployment

```
//Turning off FLEX_SPCT_ACCESS_SW for the NR TDD cell
MOD NRDUCELLCARRMGMT: NrDuCellId=0, MultiBandFlexSchSwitch=FLEX_SPCT_ACCESS_SW-0;
//Turning off FSA_CARR_SELECT_SW for NR TDD1 cell
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=FSA_CARR_SELECT_SW-0;
//((Optional) Disabling optimized fallback upon uplink CA exceptions
MOD GNODEBCARRMGMT: CaEnhancedAlgoSwitch=UL_CA_FALLBACK_DL_ONLY_CA_SW-0;
//((Optional) Disabling multi-frequency actual-PUSCH-based inner-loop power control
MOD NRDUCELLULPCONFIG: NrDuCellId=0, UIPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-0;
//Turning off FLEX_SPCT_ACCESS_SW for the NR FDD cell
MOD NRDUCELLCARRMGMT: NrDuCellId=1, MultiBandFlexSchSwitch=FLEX_SPCT_ACCESS_SW-0;
//((Optional) Disabling multi-frequency actual-PUSCH-based inner-loop power control
MOD NRDUCELLULPCONFIG: NrDuCellId=1, UIPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-0;
//Turning off FLEX_SPCT_ACCESS_SW for the SUL cell
MOD NRDUCELLCARRMGMT: NrDuCellId=2, MultiBandFlexSchSwitch=FLEX_SPCT_ACCESS_SW-0;
//((Optional) Disabling multi-frequency actual-PUSCH-based inner-loop power control
MOD NRDUCELLULPCONFIG: NrDuCellId=2, UIPwrCtrlAlgoExtSwitch=MF_ACTUAL_PUSCH_ILPC_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Use any of the following methods to check whether FSA has taken effect:

Using MML Commands

Run the **DSP NRCELLUENUMBER** command to check the value of **Flexible Spectrum Access UE Number**. If the value is not **0**, FSA has taken effect.

Observing Counters

Observe the value of the **N.User.FSA.Max** counter. If the value is greater than 0, FSA has taken effect.

Using Signaling Tracing

- Step 1** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.
- Step 2** In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **Uu Interface Trace**.
- Step 3** Check the *spCellConfigDedicated* IE in the RRCReconfiguration message. If the IE indicates that the UE's uplink carriers involve at least three frequency bands, including the SUL frequency band, FSA has taken effect.

----End

4.4.3 Network Monitoring

FSA increases the average uplink throughput of FSA UEs in a cell. As a result, the average uplink UE throughput of the entire cell increases.

The average uplink throughput of FSA UEs in a cell can be obtained using the following formula: $(N.ThpVol.FSA.UL - N.ThpVol.FSA.UL.SmallPkt) / N.ThpTime.FSA.UL.RmvSmallPkt$.

The uplink data split for FSA UEs in a cell can be observed using the **N.ThpVol.FSA.UL.Cell** counter.

[Table 4-7](#) describes these counters.

Table 4-7 FSA-related counters

Counter ID	Counter Name	Counter Description	Remarks
1911847076	N.ThpVol.FSA.UL	Total volume of uplink data received at the RLC layer for FSA UEs in a cell	This counter can be observed only in an NR TDD PCell.
1911847078	N.ThpVol.FSA.UL.SmallPkt	Volume of uplink small packets received at the RLC layer for FSA UEs in a cell	This counter can be observed only in an NR TDD PCell.
1911847082	N.ThpTime.FSA.UL.RmvSmallPkt	Uplink data transmission duration of FSA UEs excluding that of small packets in a cell	This counter can be observed only in an NR TDD PCell.

Counter ID	Counter Name	Counter Description	Remarks
1911847080	N.ThpVol.FSA.UL.Cell	Total volume of uplink data received at the MAC layer for FSA UEs in a cell	This counter can be observed in all cells involved in FSA.

 NOTE

- These counters are available only when the gNodeBs that serve the related cells run V100R021C00 or a later version.
- The **N.ThpVol.FSA.UL.Cell**, **N.ThpVol.FSA.UL**, and **N.ThpVol.FSA.UL.SmallPkt** counters can be measured only after the UE capability is reported.

The average uplink UE throughput in a cell is measured by **User Uplink Average Throughput (DU)**.

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- *Super Uplink*
- *Carrier Aggregation*
- *CA Experience Improvement*
- *VoNR*
- *FWA*
- *Ultimate Cloud X Experience*
- *LTE FDD and NR Dynamic Spectrum Sharing*
- *LTE FDD and NR Hybrid Dynamic Spectrum Sharing*
- *High Speed Mobility*
- *Hyper Cell*
- *Cell Combination*
- *Fusion Cell (Low-Frequency TDD)*
- *Positioning*
- *Cell Management*
- *Remote Interference Management (Low-Frequency TDD)*
- *Power Control*
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- 3GPP TS 38.212: "NR; Multiplexing and channel coding"
- 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"